2
Process Measurement and Transducers

2.1 Objectives

At the conclusion of this chapter, the student should:

- Be able to explain the meaning of the terms accuracy, precision, sensitivity, resolution, repeatability, rangeability, span and hysteresis;
- Be able to make an appropriate selection of sensing devices for a particular process;
- Describe the sensors used for measurement of temperature, pressure, flow, and liquid level;
- List the methods of minimizing the interference effects of noise on our instrumentation system.

2.2 The Definition of Transducers and Sensors

A transducer is a device that obtains information in the form of one or more physical quantities and converts this into an electrical output signal. Transducers consist of two principle parts, a primary measuring element referred to as a sensor, and a transmitter unit responsible for producing an electrical output that has some known relationship to the physical measurement as the basic components.

In more sophisticated units, a third element may be introduced which is quite often microprocessor based. This is introduced between the sensor and the transmitter part of the unit and has amongst other things, the function of linearizing and ranging the transducer to the required operational parameters.

2.3 Listing of Common Measured Variables

In descending order of frequency of occurrence, the principal controlled variables in process control systems comprise:

1. Temperature
2. Pressure
3. Flow rate
4. Composition
5. Liquid level

Sections 2.4 through to 2.6 of this chapter list and describe these different types of transducers; ending with a methodology of selecting sensing devices.

2.4 The Common Characteristics of Transducers

All transducers, irrespective of their measurement requirements, exhibit the same characteristics, such as Range, Span etc. This section explains and demonstrates the interpretation of the most common of these characteristics.

2.4.1 Dynamic and Static Accuracy

The very first, and most common term *Accuracy* is also the most misused-used and least understood.

It is nearly always quoted as "this instrument is +/- X% *Accurate*, when in fact it should be stated as "this instrument is +/- X% *Inaccurate*". In general, accuracy can best be described as how close the measurement's indication is to the absolute or real value of the process variable. In order to obtain a clear understanding of this term, and all of the other ones that are associated with it, the term ***Error*** should first be defined.

The Definition of Error in Process Control

Error means a mistake or transgression, and is the difference between a perfect measurement and what was actually measured, at any point, time and direction of process movement in the process measuring range.

There are two types of accuracy, *static* or *steady-state* accuracy and *dynamic* accuracy.

- *Static accuracy* is the closeness of approach to the true value of the variable when that true value is constant.
- *Dynamic accuracy* is the closeness of approach of the measurement when the true value is changing, remembering that a measurement lag occurs here, that is to say, by the time the measurement reading has been acted upon, the actual physical measured quantum may well have changed.

In addition to the term *Accuracy*, a sub-set of terms appear, these being *Precision, Sensitivity, Resolution, Repeatability and Rangeability* all of which have a relationship and association with the term ***ERROR***.

2.4.2 Precision

Precision is the accuracy with which repeated measurements of the same variable can be made under identical conditions.

In process control, precision is more important than accuracy, i.e. it is usually preferable to measure a variable precisely than it is to have a high degree of absolute accuracy. The difference between these two properties of measurement is illustrated in Figure 2.1.

Using a fluid as an example, the dashed curve represents the actual or real temperature. The upper measurement illustrates a precise but inaccurate instrument while the lower measurement illustrates an imprecise but more accurate instrument. The first instrument has the greater error, the latter has the greater drift.

(*DRIFT*: An undesirable change in the output to input relationship over a period of time)

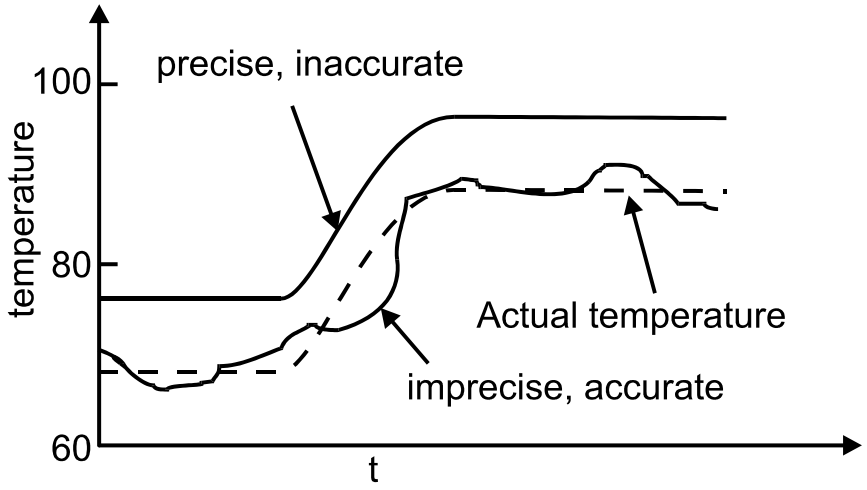


Figure 2.1
Accuracy versus Precision related to a typical Temperature Measurement

2.4.3 Sensitivity

Generally, sensitivity is defined as:

the amount of change in the output signal from a transducers transmitting element to a specified change in the input variable being measured.
i.e. it is the ratio of the output signal change to the change in the measured variable and is a steady-state ratio or the steady-state gain of the element.

So, the greater the output signal change from the transducers transmitter for a given input change, the greater the sensitivity of the measuring element.

Highly sensitive devices, such as thermistors, may change resistance by as much as 5% per ^{o}C, while devices with low sensitivity, such as thermocouples, may produce an output voltage which changes by only 5µV (5 x 10^{-6} Volts) per ^{o}C. Sensitivity.

The second kind of sensitivity important to measuring systems is defined as:

the smallest change in the measured variable which will produce a change in the output signal from the sensing element..

In many physical systems, particularly those containing levers, linkages, and mechanical parts, there is a tendency for these moving parts to stick and to have some free play.

The result of this is that small input signals may not produce any detectable output signal. To attain high sensitivity, instruments need to be well-designed and well-constructed. The control system will then have the ability to respond to small changes in the controlled variable; it is sometimes known as *resolution***.**

2.4.4 Resolution

Precision is related to *Resolution*, which is defined as the smallest change of input that results in a significant change in transducer output.

2.4.5 Repeatability

The closeness of agreement between a number of consecutive measurements of the output for the same value of input under identical operating conditions, approaching from the same direction for full range transverses. It is usually expressed as Repeatability in percent of span.
It does not include hysteresis.

2.4.6 Rangeability

This is the region between stated upper and lower range-values of which the quantity is measured. Unless otherwise stated, Input range is implied.

Example:
If the range is stated as 50°C to 320°C then the Range is quoted as 50°C to 320 ^{O}C.

2.4.7 Span

Span should not be confused with rangeability, although the same points of reference are used. Span is the ***ALGEBRAIC*** difference between the ***upper and lower range values.***

Example:
If the range is stated; as in section 2.4.6; as 50°C to 320°C then the Span is 320-50 = 270 ^{O}C.

2.4.8 Hysteresis

This is a dynamic measurement, and shows as the dependency of an output value, given for an excursion of the input, as compared with the history of prior excursions and the direction of the transverse.

Example:
If an input into a system is moved between 0% and 100% and the resultant output recorded and then the input is returned back to 0%, again with the output recorded the difference between the two values, $0\% \Rightarrow 100\% \Rightarrow 0\%$, as recorded, gives the hysteresis value of the system at all points in its range. Repetitive tests must be done under identical conditions.

2.5 Sensor Dynamics

Process dynamics have been discussed in Chapter 1, and these same factors will apply to a sensor making it important to gain an understanding of sensor dynamics. The speed of response of the primary measuring element is often one of the most important factors in the operation of a feedback controller. As process control is continuous and dynamic, the rate at which the controller is able to detect changes in the process will be critical to the overall operation of the system.

Fast sensors allow the controller to function in a timely manner, while sensors with large time constants are slow and degrade the overall operation of the feedback loop. Due to their influence on loop response, the dynamic characteristics of sensors should be considered in their selection and installation.

2.6 Selection of Sensing Devices

A number of factors must be considered before a specific means of measuring the Process Variable (PV) can be selected for a particular loop:

- The normal range over which the PV may vary, and if there are any extremes to this.
- The accuracy, precision and sensitivity required for the measurement.
- The sensor dynamics required.
- The reliability that is required.
- The costs involved, including installation and operating costs as well as purchase costs.
- The installation requirements and problems, such as size and shape restraints, remote transmission, corrosive fluids, explosive mixtures, etc.

2.7 Temperature Sensors

Temperature is the most common PV measured in process control. Due to the vast temperature range that needs to be measured (from absolute zero to thousands of degrees) with spans of just a few degrees and sensitivities down to fractions of a degree, there is a vast range of devices that can be used for temperature measurements.

The five most common sensors; *Thermocouples, Resistance Temperature Detectors or RTD's, Thermistors, IC Sensors and Radiation Pyrometers* have been selected for this chapter as they illustrate most of the application, range, accuracy and linearity aspects that are associated with temperature measurements.

2.7.1 Thermocouples

Thermocouples cover a range of temperatures, from -262°C up to +2760°C and are manufactured in many materials, are relatively cheap, have many physical forms, all of which make them a highly versatile device.

Thermocouples suffer from two major problems that cause errors when applying them to the process control environment.

- The first is the small voltages generated by them, for example a 1 ^{O}C temperature change on a platinum thermocouple results in an output change of only 5.8 μV. = (5.8 x 10^{-6} Volts).
- The second is their nonlinearity, requiring polynomial conversion, look up tables or related calibration to be applied to the signaling and controlling unit. See Figure 2.2.

Ranges of Six types of Common Thermocouples

Metal Composition		*Temperature Span*	*Seebeck Coefficient*
K	*Chromel vv Alumel*	*-190 to +1371 ^{O}C*	*40 μV/^{O}C*
J	*Iron vv Constantan*	*-190 to +760 ^{O}C*	*50 μV/^{O}C*
T	*Copper vv Constantan*	*-190 to +760 ^{O}C*	*50 μV/^{O}C*
E	*Chromel vv Constantan*	*-190 to +1472 ^{O}C*	*60 μV/^{O}C*
S	*Platinum vv 10% Rhodium/Platinum*	*0 to +1760 ^{O}C*	*10 μV/^{O}C*
R	*Platinum vv 13% Rhodium/Platinum*	*0 to +1670 ^{O}C*	*11 μV/^{O}C*

Table 2.1
Thermocouple Types, Temperature Range and Value of the Seebeck Effect

Principles of Thermocouple Operation

A thermocouple could be considered as a heat operated battery, consisting of two different types of homogeneous (of the same kind and nature) metal or alloy wires joined together at one end, the measuring point and connected, usually via special compensating cable, to some form of measuring instrument. At the point of connection to the measuring device a second junction is formed, called the reference or cold junction, this completes the circuit.

The Peltier & Thomson Effects on Thermocouple Operation

The *PELTIER EFFECT* is the cause of the emf's generated at every junction of dissimilar metals in the circuit. This effect involves the generation or absorption of heat at the junction as current flows through it and temperature is dependent on current flow direction.

The *THOMSON EFFECT*, where a second emf can also be generated along the temperature gradient of a single homogeneous wire can also contribute to measurement errors. It is essential that all the wire in a thermocouple measuring circuit is homogeneous as then the emf's generated will be dependant solely on the types of material used. Any thermal emf's generated in the wire when it passes through temperature gradients will also be cancelled from one to the other.

Additionally, if both junctions of a homogeneous metal are held at the same temperature, the metal will not contribute additional emf's to the circuit. It follows then that if all junctions in the circuit are held at a constant temperature, except the measuring one, measurement can be made of the hot, or measuring, junction value against the constant value or cold junction reference value.

Reference or Cold Junction Compensation

As described in section 2.7.1.3, we have to ensure that all the junctions in the measuring circuit, with the exception of the one being used for the actual process measurement, must either:

- Be held at a constant known temperature, usually $0^{O}C$, and called a "Cold Junction".
- Or the temperature of these junctions should be measured and the measuring instrument take this into consideration when calculating its final output.

Both methods are both commonly used, the first one, the cold junction, utilizes an Isothermal block held at a known temperature and in which the connections from the thermocouple wires to copper wires is made. The second method is to measure the temperature, usually by a thermistor, at the point of copper to thermocouple connections, feed this value into the measuring system and have that calculate a corrected output.

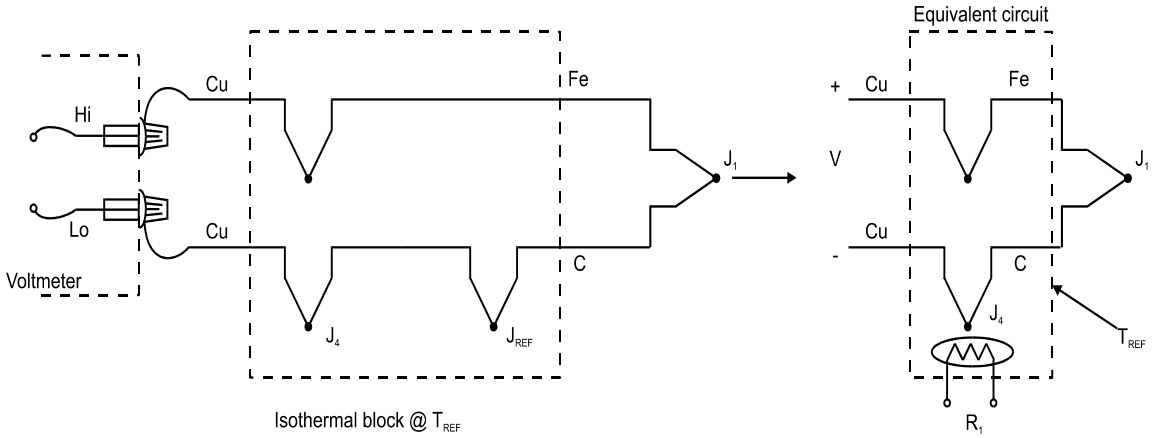


Figure 2.2
Thermocouple Cold Junction and Reference Junction Circuit Examples

2.7.2 Resistance Temperature Detectors or RTD's

In the same year as the discovery of the thermocouple by Thomas Seebeck, Sir Humphrey Davy noted the temperature / Resistivity dependence of metals, but it was H C Meyers who developed the first RTD in 1932.

Construction of RTD's

RTD's consist of a Platinum or Nickel wire element encased in a protective housing having, in the case of the Platinum version a base resistance of 100 Ohms at 0°C and the Nickel type a resistance of 1000 ohms, again at 0°C.

They come packaged in either 2,3 or 4 wire versions, the 3 and 4 wire being the most common. 2 wire versions can be very inaccurate as the lead resistance is in series with the measuring circuit, and the measuring element relies on resistance change to indicate the temperature change.

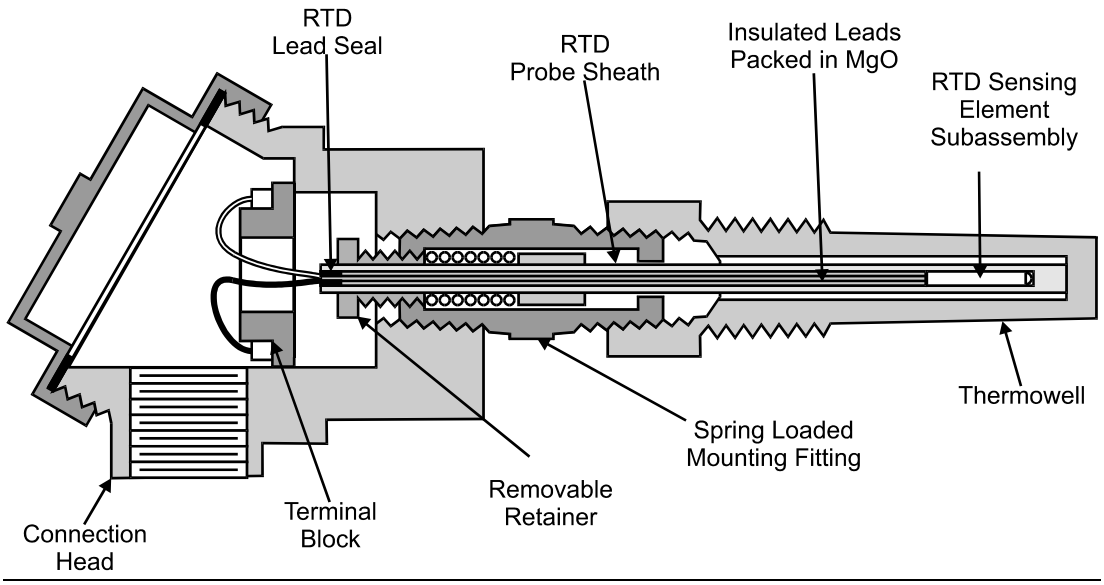


Figure 2.3
Construction of RTD

Range Sensitivity and Spans of RTD's

RTD's operate over a narrower range than thermocouples, from -247 ^{O}C to + 649 ^{O}C. Span selection has to be made for correct operation as typically the sensitivity of a PT100 is 0.358 Ohms / ^{O}C about the nominal resistance of 100 Ohms at 0°C.

This corresponds to a single resistance range of (100-88 Ohms = -247 ^{O}C to 100+232 Ohms = 649 ^{O}C) resulting in 12 to 332 Ohms, which is outside the range of a single transducer.

Example of RTD Application in a Digital Environment

Figure 2.3 shows the configuration of a 3 wire RTD used in a digital process control application.

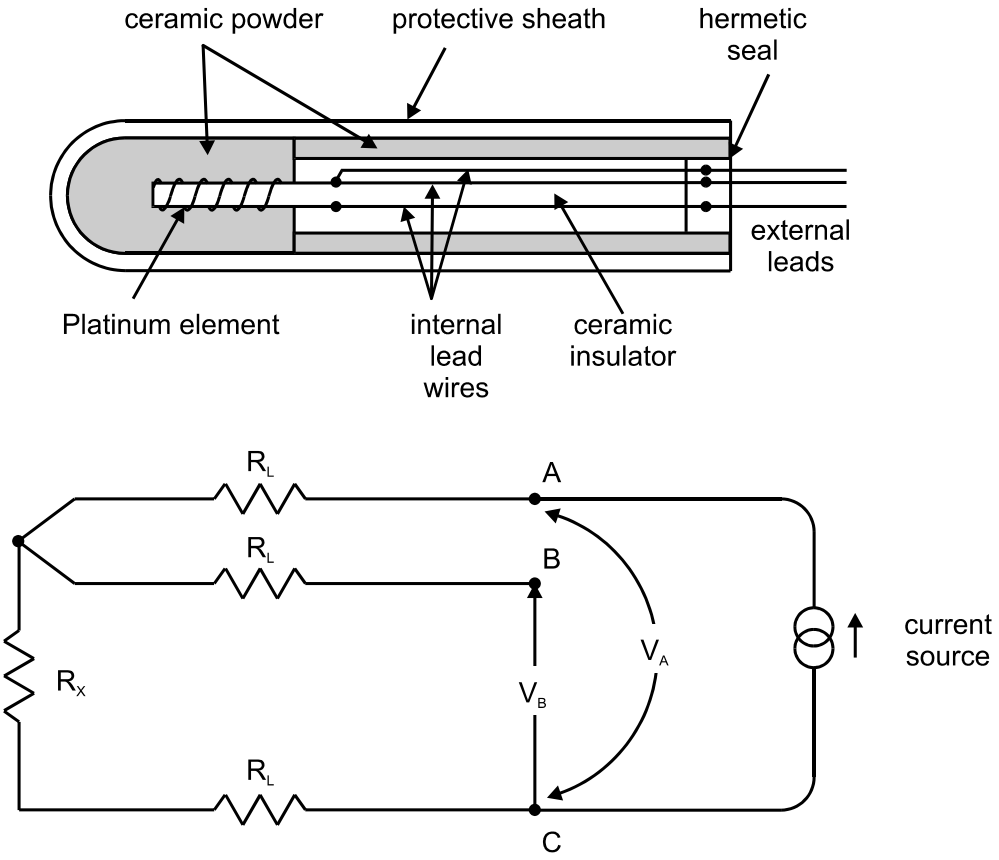


Figure 2.4
3-Wire RTD Configuration for a Digital System

Modern digital controllers use these 3-wire RTD's in the following manner:

A constant current generator drives a current through the circuit [A to C] consisting of $2R_L + R_X$. A voltage detector reads a voltage, V_B, proportional to $R_X + R_L$ between points [B and C] and a second voltage V_A which is proportional to $R_X + 2R_L$ between points [A and C].

As $V_A - V_B$ is proportional to R_L so $V_B - V_A - V_B$ is proportional to R_X alone where:

- R_L = The resistance of each of the three RTD's leads.
- R_X = The measuring element of the RTD.
- V_A = The voltage supply to the RTD's measuring element R_X from the constant current source.
- V_B = The final measured voltage, or output from the RTD (3 wire version).

The measurements are made sequentially, digitized and stored until differences can be computed.

RTD's are reasonably linear in operation, see Figure 2.6, but this depends to a great extent on the area of operation being used within the total span of the particular transducer in question.

Self Heating Problems Associated with RTD's

RTD's suffer also from an effect of self-heating, where the excitation current heats the sensing element, thereby causing an error, or temperature offset. Modern digital systems can overcome this problem by energizing the transducer just before a reading is taken. Alternatively the excitation current can be reduced but this is at the expense of lower measuring voltages occurring across the transducers output, and subsequently induced electrical noise can become a problem. Lastly the error caused by self-heating can be calculated and adjustment made to the measuring algorithms.

2.7.3 Thermistors

These elements are the most sensitive and fastest temperature measuring devices in common use, unfortunately the price paid for this is terrible non-linearity (see Figure 2.6), and a very small temperature range.

Thermistors are manufactured from metallic oxides, and have a negative temperature coefficient, that is their resistance drops with temperature rise. They are also manufactured in almost any shape and size from a pin head to discs up to 25 mm diameter x 5 mm thick.

Thermistor Values, Range and Sensitivity

Most thermistors have a nominal quoted resistance of about 5000 ohms and because of their sensitivity, this base resistance is quoted at a specific temperature; reference having to be made to the relative type in the manufacturer's published specifications.

Thermistor values can change by as much as 200 ohms per ^{O}C which, in this case would give a maximum range of only +25 degrees from the quoted base temperature.

2.7.4 I.C Sensors

Integrated Circuitry Sensors have only recently began to make their presence felt in the process control world. As such they are still limited in the variability of shape, size and packaging that is advisable.

Their main advantages are their low cost (below $10.00) along with their linear and high output signals.

IC Sensor Ranges and Accuracy

As these sensors are formed from integrated silicon chips, their range is limited to -55°C to +150°C but easily have calibrated accuracies to 0.05 ^{O}C to 0.1^{O}C.

Cryogenic Temperature Measurements

An exception to the normal operating temperature range of IC Sensors is a version that can be used for cryogenic temperatures -271°C to +202°C by the application of special diodes designed exclusively to operate at these sub-normal temperatures (absolute zero = -273.16 ^{O}C).

2.7.5 Selection of Temperature Transducer Design and Thermowells

Temperature measurement transducers, in particular thermocouples; need different housings and mountings depending on the application requirements.

Sensing devices are usually mounted in a sealed tube, more commonly known as a thermowell; this has the added advantages of allowing the removal or replacement of the sensing device without opening up the process tank or piping. Thermowells need to be considered when installing temperature-sensing equipment. The length of the thermowell needs to be sized for the temperature probe.

Consideration of the thermal response needs to be taken into account. If a fast response is required, and the sensor probe already has adequate protection, then a thermowell may impede system performance and response time. Note that when a thermowell is used, the response time is typically doubled.

Thermowells can provide added protection to the sensing equipment, and can also assist in maintenance and period calibration of equipment.

Thermopaste assists in the fast and effective transfer of thermal dynamics from the process to the sensing element. Application and maintenance of this material needs to be considered. Regular maintenance and condensation can affect the operation of the paste.

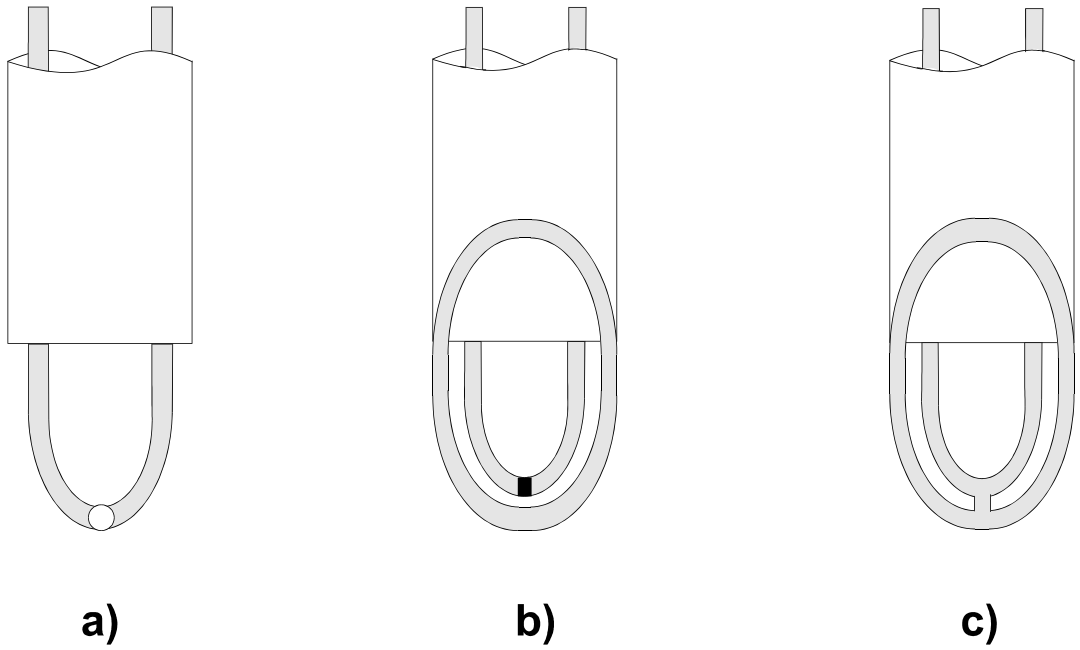


Figure 2.5
Sectional views of three typical thermocouple probes

Figure 2-5 shows the three typical designs of thermocouple probes:

- Open Ended; Subject to damage and should not be used in a hostile environment.
- Sealed and both thermally and electrically isolated from the outside world.
- Sealed but with thermal (and / or electrical) connection to the outside world.

2.7.6 Radiation Pyrometers

At the other end of the scale is the requirement to measure high temperatures up to 4000°C or more. Total radiation pyrometers operate by measuring the total amount of energy radiated by a hot body. Their temperature range is 0°C to 3890°C.

The infrared (IR) pyrometer is rapidly replacing this older type of measurement, and these work by measuring the dominant wavelength radiated by a hot body. The basis of this is in the fact that as temperature increases the dominant wavelength of hot body radiation gets shorter.

Developments in Infrared Optical Pyrometry

Two recent developments in the world of Pyrometry that should be mentioned are the utilization of lasers and fiber optics.

Lasers are used to automatically correct errors occurring due to changes in surface emissivity as the object's temperature changes.

Fiber optics can focus the temperature measurements on inaccessible or unfriendly areas. Some of these units are capable of very high accuracy, typically 0.1% @ 1000°C and can operate from 500°C up to 2000°C. Multiplexing of the optics is also possible, reducing costs in multi-measuring environments.

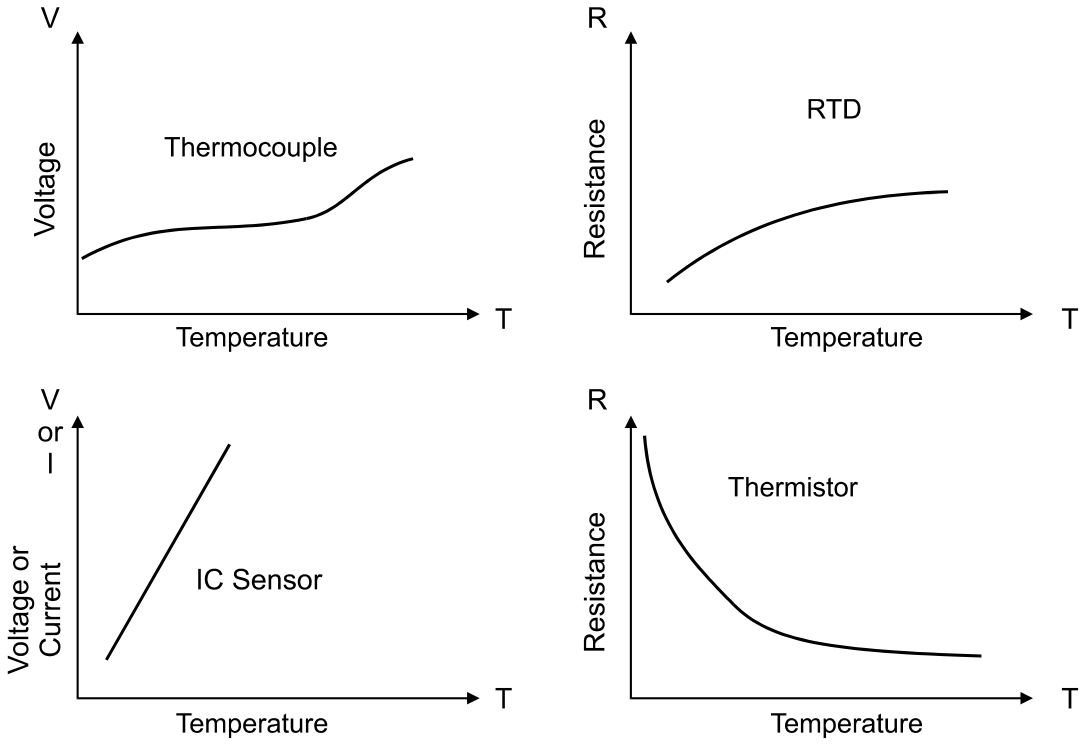


Figure 2.6
Characteristics of Thermocouples, RTD's IC and Thermistor Temperature Sensors

2.8 Pressure Transmitters

Pressure is probably the second most commonly used and important measurement in process control. The most familiar pressure measuring devices are manometers and dial gauges, but these require a manual operator.

For use in process control, a pressure-measuring device needs a pressure transmitter that will produce an output signal for transmission, e.g. an electric current proportional to the pressure being measured. A transmitter typically produces an output of a 4-20 mA signal, is rugged, and can be used in flammable or hazardous service.

2.8.1 Terms of Pressure Reference

Pressure is defined as force per unit area, and may be expressed in units of newtons per square meter, millimeters of mercury, atmospheres, bars or torrs.

There are three common references against which it can be measured:

- If measured against a vacuum, the measured pressure is called absolute pressure.
- Against local ambient pressure it is gauge pressure.
- If the reference pressure is user supplied, differential pressure is measured.

There are seven principle methods of electronically measuring pressure for use in process control and each of these is listed and described under its numeric heading, in principle detail below:

- 2.8.2 Strain Gauge (Bonded or Un-bonded wire or foil, bonded or diffused semiconductor)
- 2.8.3 Capacitive
- 2.8.4 Potentiometric
- 2.8.5 Resonant wire
- 2.8.6 Piezoelectric
- 2.8.7 Magnetic (Inductive and Reluctive)
- 2.8.8 Optical

2.8.2 Strain Gauge

In process control applications, one of the most common ways to measure pressure is using a strain gauge sensor. There are two basic types of strain gauge, bonded and unbonded, each utilizing wire or foil, but both working in the same electrical manner. A thin wire (or foil strip), usually made from Chrome-Nickel alloys and sometimes Platinum, is subjected to stretching, and hence its resistance increases as its length increases.

Strain gauges are commonly made using a thin metal wire or foil that is only a few micrometers thick, so they can also be known as thin film based bonded pressure sensors.

The unbonded strain gauge consists of open wire wound round two parallel mounted posts which are flexed or pulled apart, hence imparting a stretching dynamic to the wire, reducing its resistance, these being physically much larger units. See Figure 2.7.

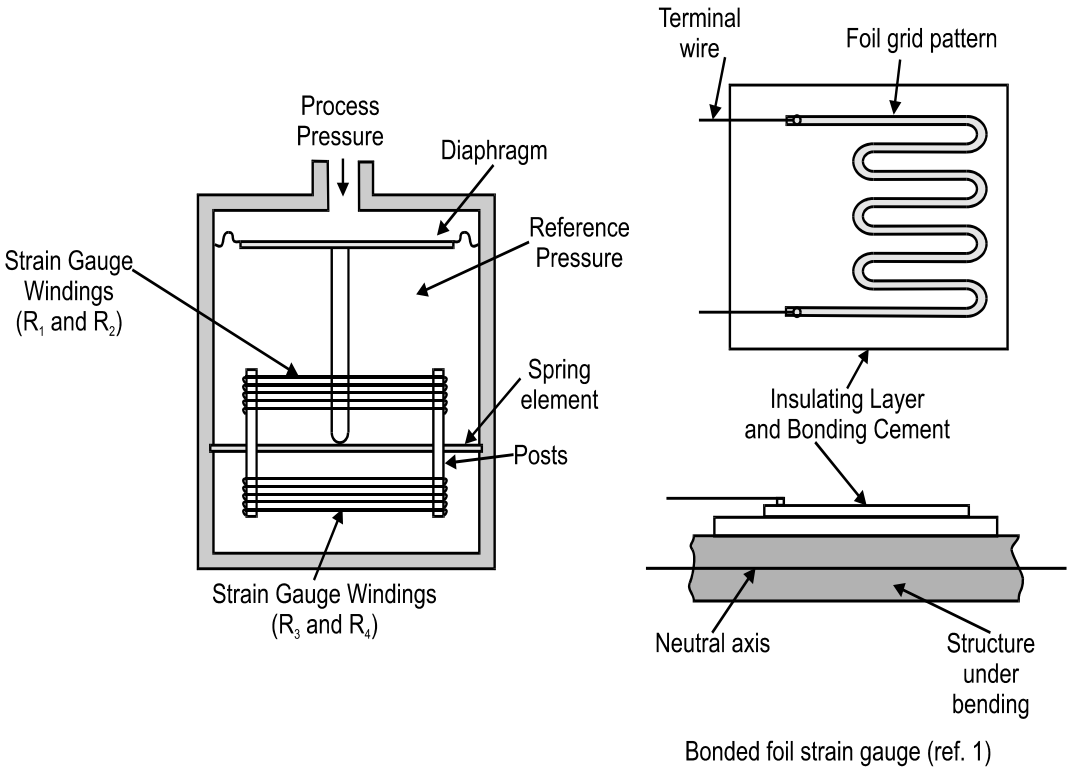


Figure 2.7
Composition of Bonded and Unbonded strain gauges

Composition of Strain Gauges

Bonded strain gauges are the most common type in use. They comprise an insulated bonded sandwich usually made of two sheets of paper, with the gauge wire laid in a specific pattern between them. Strain gauge wires of less than 0.001 in. (0.025 mm.) diameter are used as they have a surface area several thousand times greater than the cross sectional area.

Foil gauges have been commercially made where the foil thickness can be as low as 0.0001 in. (0.025 mm).

Semiconductor types are available which have sensitivities close to one hundred times greater than the wire types.

Strain Gauge Sensitivity and Gauge Factor

The ratio of the percentage change in resistance to the percentage change in length is a measure of the sensitivity (S) of the gauge:

$$S = \frac{\Delta R / R}{\Delta L / L}$$

where:

L is the initial length of the wire or foil.
R is the specific resistivity in the unstrained position.

Many things effect the axial strain and gauge resistance, such as the geometry of the wire or foil in the gauge, direction of strain, This is expressed by the constant called Gauge Factor or GF:

$$GF = \frac{\Delta R / R}{Axial \text{ Strain}}$$

Typically, four strain gauges are bonded to a metal or plastic flexible diaphragm and connected into a Wheatstone bridge circuit to yield an electrical signal proportional to the strain caused by the displacement of the diaphragm by the pressure applied to it.

The changes of resistance in a strain gauge are very small and as such precise and accurate instrumentation is required in order to obtain useable and accurate readings. The most common form of measuring is in a Wheatstone bridge circuit. Figure 2.8 shows a typical arrangement using this type of instrument.

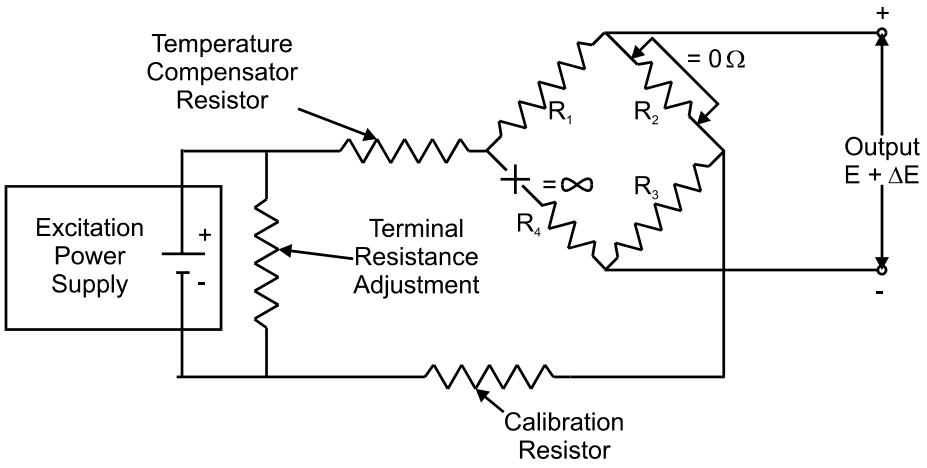


Figure 2.8
Wheatstone Circuit for Strain Gauge Measurement

Effects of temperature *(ref. Section 2.7.2.3)* on the gauge's resistance is minimal due to their influence on the output being subtractive.

The active gauges are located on opposite arms of the bridge making effects additive, the other 2 gauges are for compensation and are either “dummy” gauges or resistors with equal resistance to the active gauges.

This circuit is suited to both static and dynamic strain measurements, the output is, however, a differential output and care must be exercised in “grounding” any part of this circuit.

An alternative arrangement, called a ballast or potentiometric circuit is arrived at by removing R2 and making the value of R_3 equal to 0 as indicated by the dotted lines in Figure 2.8.

Unfortunately this circuit is best used for dynamic sensing only, however one of the signal leads can now be grounded or electrically referenced to 0 volts.

Measurement Errors For Strain Gauges

A number of errors exist when strain is measured using the Wheatstone bridge arrangement. Typical these are:

- Gauge factor uncertainty (typically 1%).
- Bridge non-linearity (typically 1%). This is a result of the assumption that the change in strain gauge resistance is very small compared to the nominal gauge resistance.
- Matching of compensation resistors to the strain gauge (typically 0.5%).
- Measurement errors caused by the accuracy and resolution of the measuring device and lead resistances.
- Temperature effects. Resistance varies with changes from the temperature at which a bridge is calibrated.
- Self heating of gauges.

Strain Gauge Pressure Transducer Specifications

Strain gauge elements can detect absolute gauge, and differential spans from 30 in. H_2O to upwards of 200,000 psig (7.5 kPa to 1400MPa). Their inaccuracy is around 0.2% and 0.5% of Span. Units are available to work in the temperature range of -50°C to +120°C with special units going to 320°C.

2.8.3 Vibrating Wire or Resonant Wire Transducers

This type of sensor consists of an electronic oscillator circuit, which causes a wire to vibrate at its natural frequency when under tension. The principle is similar to that of a guitar string. A thin wire inside the sensor is kept in tension, with one end fixed and the other attached to a diaphragm. As pressure moves the diaphragm, the tension on the wire changes thereby changing its resonant vibration frequency.

These frequency changes are a direct consequence of pressure changes and as such is detected and shown as pressure.

The frequency can be sensed as digital pulses from an electromagnetic pickup or sensing coil. An electronic transmitter would then convert this into an electrical signal suitable for transmission.

This type of pressure measurement can be used for differential, absolute or gauge installations. Absolute pressure measurement is achieved by

evacuating the low-pressure diaphragm. A typical vacuum pressure for such a case would be about 0.5 Pa.

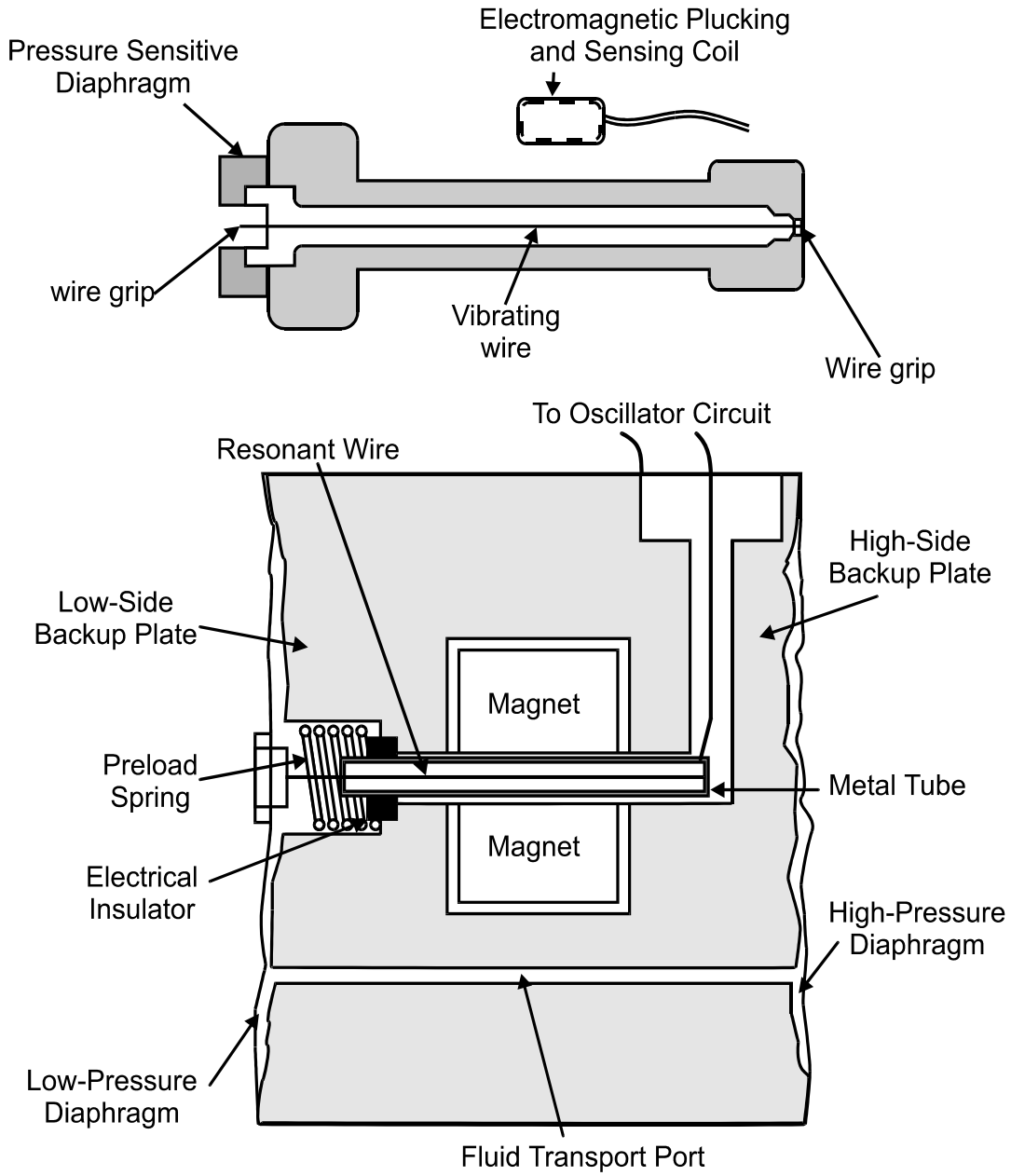


Fig 2.9
Resonant Wire Transducer

Transducer Advantages

- Good accuracy and Repeatability
- Stable
- Low hysteresis
- High-resolution

- Absolute, gauge or differential measurement

Transducer Limitations, or Disabilities

- Sensitivity to vibrations
- Temperature variations require temperature compensation within the sensor, this problem limits the sensitivity of the device
- The output generated is non-linear which can cause continuous control problems
- This technology is seldom used any more. Being older technology it is typically found with analog control circuitry.

2.8.4 Capacitance

Capacitive pressure measurement involves sensing the change in capacitance that results from the movement of a diaphragm. The sensor is energized electrically with a high frequency oscillator. As the diaphragm is deflected due to pressure changes, the relative capacitance is measured by a bridge circuit.

Two designs are quite common. The first is the two-plate design and is configured to operate in the balanced or unbalanced mode. The other is a single capacitor design.

The balanced mode is where the reference capacitor is varied to give zero voltage on the output. The unbalanced mode requires measuring the ratio of output to excitation voltage to determine pressure.

This type of pressure measurement is quite accurate and has a wide operating range. Capacitive pressure measurement is also quite common for determining the level in a tank or vessel.

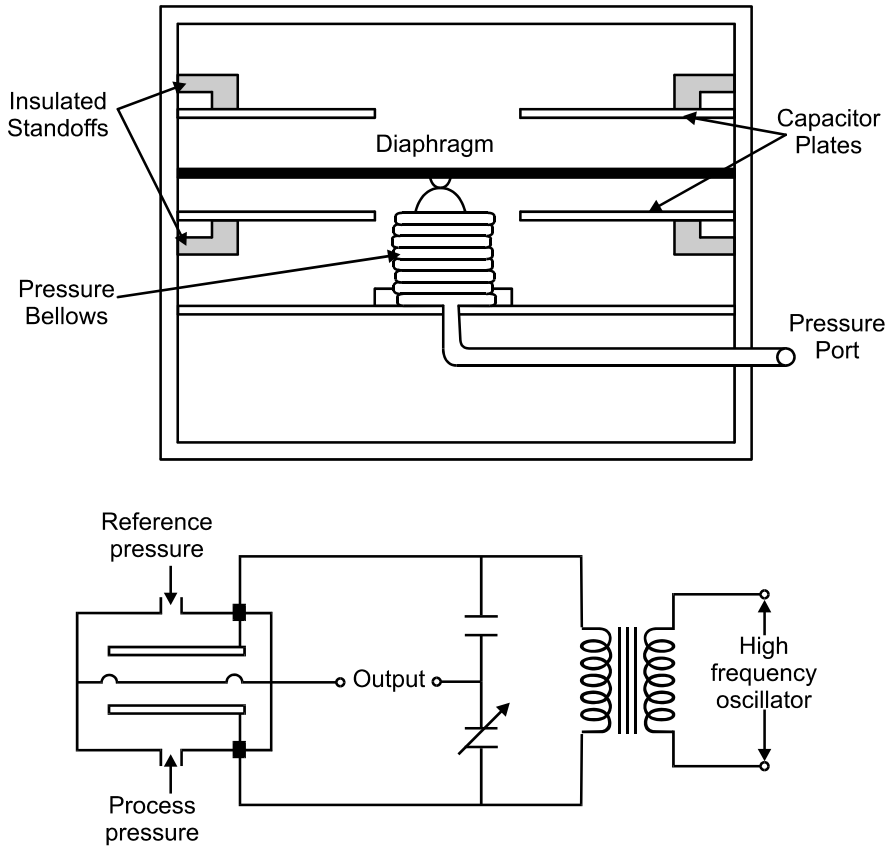


Fig 2.10
Capacitance Pressure Detector

Transducer Advantages

- Inaccuracy 0.01 to 0.2%
- Range of 80Pa to 35MPa
- Linearity
- Fast response

Transducer Limitations

- Temperature sensitive
- Stray capacitance problems
- Vibration
- Limited over pressure capability
- Cost

Many of the disadvantages above have been addressed and their problems reduced in newer designs.

Temperature controlled sensors are available for applications requiring a high accuracy.

With strain gauges being the most popular form of pressure measurement, capacitance sensors are the next most common solution.

2.8.5 Linear Variable Differential Transformer

This type of pressure measurement relies on the movement of a high permeability core within transformer coils. The movement is transferred from the process medium to the core by use of a diaphragm, bellows or bourdon tube.

The LVDT operates on the inductance ratio between the coils. Three coils are wound onto the same insulating tube containing the high permeability iron core. The primary coil is located between the two secondary coils and is energized with an alternating current.

Equal voltages are induced in the secondary coils if the core is in the center. The voltages are induced by the magnetic flux. When the core is moved from the center position, the result of the voltages in the secondary windings will be different. The secondary coils are usually wired in series.

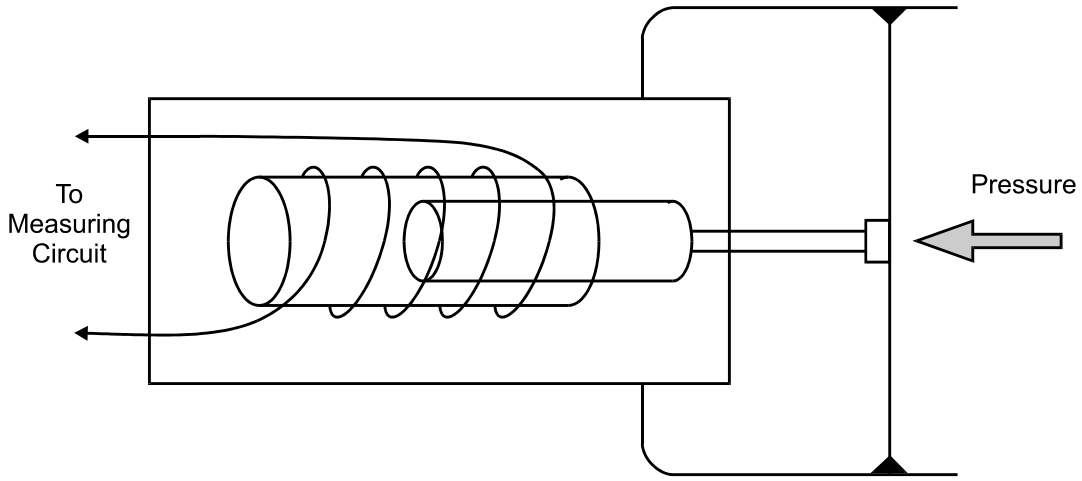


Fig 2.11
Schematic representation of a linear motion variable inductance prior transducer element (LMVIPTE)

Transducer Limitations or Disabilities

- Mechanical wear
- Vibration

Summary

- This is an older technology, used before strain gauges were developed.
- Typically found on old weigh-frames or may be used for position control applications.
- Very seldom used anymore; strain gauge types have superseded these transducers in most applications.

2.8.6 Optical

Optical sensors can be used to measure the movement of a diaphragm due to pressure. An opaque vane is mounted onto a diaphragm and moves in front of an infrared light beam. As the light is disturbed, the received light on the measuring diode indicates the position of the diaphragm.

A reference diode is used to compensate for the aging of the light source. Also, by using a reference diode, the temperature effects are cancelled as they affect the sensing and reference diodes in the same way.

Transducer Advantages

- Temperature corrected
- Good Repeatability
- Negligible hysteresis as optical sensors require very little movement for accurate sensing
- Optical pressure measurement provides very good repeatability with negligible hysteresis

Transducer Limitations or Disabilities

- Expensive

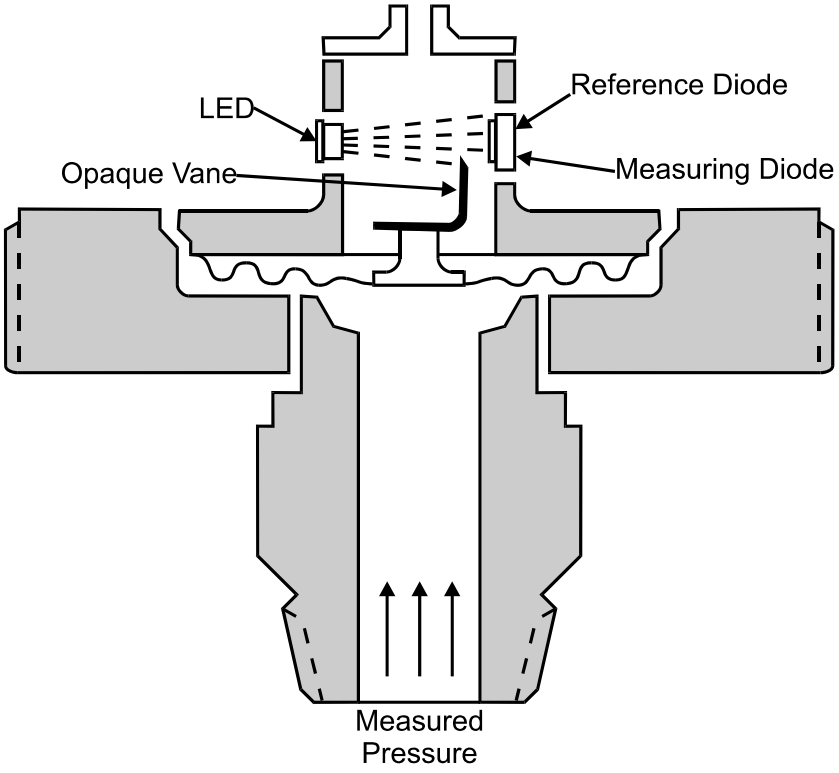


Fig 2.12
Optical Pressure Transducer

2.8.7 Pressure Measurement Applications

There are a number of requirements that need to be considered with applications in pressure measurement. Some of the more important of these are listed below:

1. Location of process connections
2. Isolation valves
3. Use of impulse tubing
4. Test and drain valves
5. Sensor construction
6. Temperature effects
7. Remote diaphragm seals
8. Corrosion may cause a problem to the transmitter and pressure sensing element
9. The sensing fluid contains suspended solids or is sufficiently viscous to clog the piping

10. The process temperature is outside of the normal operating range of the transmitter
11. The process fluid may freeze or solidify in the transmitter or impulse piping
12. The process medium needs to be flushed out of the process connections when changing batches
13. Maintaining sanitary or aseptic conditions
14. Eliminating the maintenance required with wet leg applications
15. Making density or other measurements

2.9 Flow Meters

In many industrial applications it is convenient and useful to measure flow and so a large percentage of transmitter sales are for measuring flow. As a result, there is a huge range of flowmeters to suit a variety of applications. The operation of these may conform to one of two approaches.

2.9.1 Energy Extractive Flowmeters

This is the older of the two approaches, and uses flow-measurement devices that reduce the energy of the system. The most common of these are the differential pressure producing flowmeters, such as the orifice plate, flow nozzle, and venturi tube.

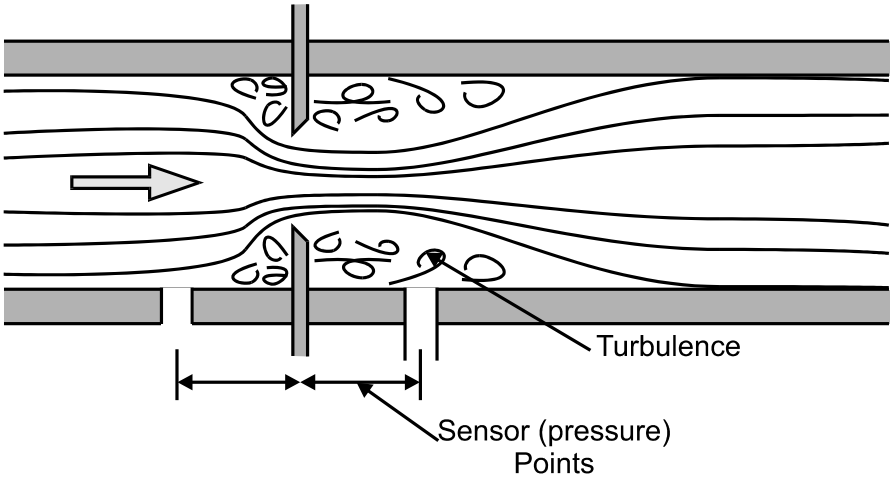


Figure 2.13
Flow Patterns with an Orifice Plate

Orifice Plate

A standard orifice plate is simply a smooth disc with a round, sharp-edged inflow aperture and mounting rings. In the case of viscous liquids, the upstream edge of the bore can be rounded. The shape of the opening and its location do vary widely, and this is dependent on the material being measured. Most common are concentric orifice plates with a round opening in the center. They produce the best results in turbulent flows when used with clean liquids and gases.

When measuring liquids the bore can be positioned at the top of the pipeline to allow the passage of gases. The same applies when allowing suspended solids to pass by positioning the bore at the bottom and gaining a more accurate liquid flow measurement.

Standard orifice meters are primarily used to measure gas and vapor flow. Measurement is relatively accurate; however because of the obstruction of flow there is a relatively high residual permanent pressure loss. They are well understood, rugged and relatively inexpensive for large pipe sizes and are suited for most clean fluids and aren't influenced by high temperatures.

Transducer Advantages

- Simple construction
- Inexpensive
- Easily fitted between flanges
- No moving parts
- Large range of sizes and opening ratios
- Suitable for most gases and liquids
- Well understood and proven
- Price does not increase dramatically with size

Transducer Limitations or Disabilities

- Inaccuracy, typically 1%
- Low rangeability, typically 4:1
- Accuracy is affected by density, pressure and viscosity fluctuations
- Erosion and physical damage to the restriction affects measurement accuracy
- Cause some unrecoverable pressure loss
- Viscosity limits measuring range
- Require straight pipe runs to ensure accuracy is maintained
- Pipeline must be full (typically for liquids)
- The inaccuracy with orifice-type measurement is due mainly to process conditions and temperature and pressure variations
- They are also affected by ambient conditions and upstream and downstream piping, as this affects the pressure and continuity of flow

Turbine or Rotor Flow Transducer

Turbine meters have rotor-mounted blades that rotate when a fluid pushes against them. They work on the reverse concept to a propeller system. In a propeller system, the propeller drives the flow; in this case the flow drives and rotates the propeller. Since it is no longer propelling the fluid, its now called a turbine.

The rotational speed of the turbine is proportional to the velocity of the fluid.

Different methods are used to convey rotational speed information. The usual method is by electrical means where a magnetic pick-up or inductive proximity switch detects the rotor blades as they turn. As each blade tip on the rotor passes the coil it changes the flux and produces a pulse. The rate of pulses indicates the flow rate through the pipe.

Turbine meters require a good laminar flow. In fact 10 pipe diameters of straight line upstream, and no less than 5 pipe diameters downstream from the meter are required. They are therefore not accurate with swirling flows.

Turbine meters are specified with minimum and maximum linear flow rates that ensure the response is linear and the other specifications are met. For good rangeability, it is recommended that the meter be sized such that the maximum flow rate of the application be about 70 to 80% of that of the meter.

Density changes have little effect on the meter's calibration.

Transducer Advantages

- High accuracy, repeatability and rangeability for a defined viscosity and measuring range
- Temperature range of fluid measurement: -220°C to +350°C
- Very high pressure capability: 9300psi
- Measurement of non-conductive liquids
- Capability of heating measuring device
- Suitable for very low flow rates

Transducer Limitations or Disabilities

- Not suitable for high viscous fluids
- Viscosity must be known
- 10D upstream and 5D downstream of straight pipe is required
- Not effective with swirling fluids
- Only suitable for clean liquids and gases
- Pipe system must not vibrate
- Specifications critical for measuring range and viscosity
- As turbine meters rely on the flow, they do absorb some pressure from the flow to propel the turbine
- The pressure drop is typically around 20 to 30 kPa at the maximum flow rate and does vary depending on flow rate
- It is a requirement in operating turbine meters that sufficient line pressure be maintained to prevent liquid cavitation
- The minimum pressure occurs at the rotor, however the pressure recovers substantially are the turbine
- If the back-pressure is not sufficient, then it should be increased or a larger meter chosen to operate in a lower operating range - this does have the limitation of reducing the meter flow range and accuracy

Summary

Turbine meters provide excellent accuracy, repeatability and rangeability for a defined viscosity and measuring range, and are commonly used for custody transfer applications of clean liquids and gases.

2.9.2 Energy Additive Flow Meters

A common example of the energy additive approach is the magnetic flowmeter, illustrated in Figure 2.14. This device is used to make flow measurements on a conductive liquid. A charged particle moving through the magnetic field produces a voltage proportional to the velocity of the particle. A conductive liquid consisting of charged particles will then produce a voltage proportional to the volumetric flow rate.

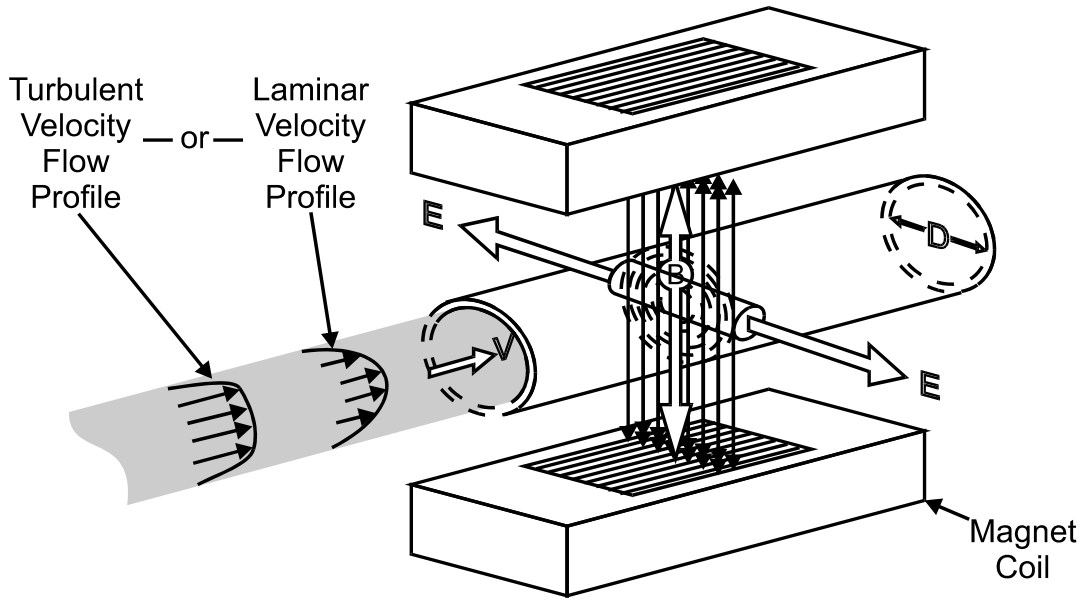


Figure 2.14
Schematic Representation of a Magnetic Flow Meter

The Magmeter

The advantages of magnetic flowmeters are:

- They have no obstructions or restrictions to flow
- No pressure drop or differential
- No moving parts to wear out
- They can accommodate solids in suspension
- No pressure sensing points to block up
- They measure volume rate at the flowing temperature independent of the effects of viscosity, density, pressure or turbulence
- Another advantage is that many magmeters are capable of measuring flow in either direction

Most industrial liquids can be measured by magnetic flowmeters, these include acids, bases, water, and aqueous solutions. However some exceptions are most organic chemicals and refinery products which have insufficient conductivity for measurement. Also pure substances, hydrocarbons and gases cannot be measured.

In general the pipeline must be full, although with newer models, level sensing takes this factor into account when calculating a flow-rate.

Accuracy

Magnetic flowmeters are very accurate and have a linear relationship between the output and flow-rate. Alternatively, the flow rate can be transmitted as a pulse per unit of volume or time.

The accuracy of most magnetic flowmeter systems is 1% of full-scale measurement. This takes into account both the meter itself and the secondary instrument. Because of its linearity, the accuracy at low flow rates exceeds that of such devices as the Venturi tube. The magnetic flowmeter can be calibrated to an accuracy of 0.5% of full scale and is linear throughout.

Selection, Sizing And Liners

Sizing of magmeters is done from manufacturer's nomographs to determine suitable diameter meters for flow rates.

The principle of operation of the magmeter requires the generation of a magnetic field, and the detection of the voltage across the flow.

If the pipe is made of a material with magnetic properties, then this will disrupt the magnetic field and effectively short circuits the magnetic field. Likewise if the inside of the pipe is conductive, then this will short circuit the electrodes used to detect the voltage across the flow.

The meter piping must be manufactured from a non-magnetic material such as stainless steel in order to prevent short-circuiting of the magnetic field.

The lining of the meter piping must also be lined with an insulating material to prevent short-circuiting of the electric field.

The liner has to be chosen to suit the application, particularly the resistance it has to the following:

- Chemical Corrosion
- Erosion
- Abrasion
- Pressure
- Temperature

Liner Materials

Teflon (PTFE)

- Widely used due to its high temperature rating
- Anti-stick properties reduce problems with build-up
- Approved for food and beverage environments
- Resistant to many acids and bases

Neoprene

- Good abrasion resistance
- Good chemical resistance

Soft Rubber

- Relatively inexpensive
- High resistance to abrasion
- Used mainly for slurry applications

Hard rubber

- Inexpensive
- General-purpose applications
- Used mainly for water and soft slurries

Ceramic

- High abrasion resistance
- High corrosion resistance
- High temperature rating
- Less expensive to manufacture
- Also suited to sanitary applications
- Strong compressive strength, but poor tensile strength
- Brittle
- May crack with sudden temperature changes, especially downward
- Cannot be used with oxidizing acids or hot concentrated caustic

Installation Techniques

For correct operation of the magmeter, the pipeline must be full. This is generally done by maintaining sufficient back-pressure from downstream piping and equipment. Meters are available that make allowance for this problem, but are more expensive and are specialized. This is mainly a problem in gravity feed systems.

Magmeters are not greatly affected by the profile of the flow, and are not affected by viscosity or the consistency of the liquid. It is however recommended that the meter be installed with 5 diameters of straight pipe upstream and 3 diameters of straight pipe downstream from the meter.

Applications requiring reduction in the pipe diameter for the meter installation need to allow for the extra length of reducing pipe. It is also recommended that in those applications that the reducing angle not be greater than 8°, although manufacturers data should be sought.

Grounding is another important aspect when installing magmeters, and manufacturers recommendations should be adhered to. Such recommendations would require the use of copper braid between the meter flange and pipe flange at both ends of the meter. These connections provide a path for stray currents and should also be grounded to a suitable grounding point. Magmeters with built in grounding electrodes eliminate this problem, as the grounding electrode is connected to the supply ground.

Transducer Advantages

- No restrictions to flow
- No pressure loss
- No moving parts
- Good resistance to erosion
- Independent of viscosity, density, pressure and turbulence
- Good accuracy
- Bi-directional
- Large range of flow rates and diameters

Transducer Limitations or Disabilities

- Expensive
- Most require a full pipeline
- Limited to conductive liquids
- As mentioned earlier, a magnetic flowmeter consists of either a lined metal tube, usually stainless steel because of its magnetic properties, or an unlined non metallic tube. The problem can arise if the insulating liners and electrodes of the magnetic flowmeter become coated with conductive residues deposited by the flowing fluid.
- Erroneous voltages can be sensed if the lining becomes conductive.
- Maintaining high flow rates reduces the chances of this happening. However, some manufacturers do provide magmeters with built in electrode cleaners.
- Block valves are used on either side of AC-type magmeters to produces zero flow and maintain full pipe to periodically check the zero. DC units do not have this requirement.

Ultrasonic flow measurement

There are two types of ultrasonic flow measurement:

- Transit time measurement, used for clean fluids
- Doppler effect, used for dirty, slurry type flows

Transit time ultrasonic flow measurement

The transit-time flowmeter device sends pulses of ultrasonic energy diagonally across the pipe. The transit-time is measured from when the transmitter sends the pulse to when the receiver detects the pulse.

Each location contains a transmitter and receiver. The pulses are sent alternatively upstream and downstream and the velocity of the flow is calculated from the time difference between the two directions.

Transit-time ultrasonic flow measurement is suited for clean fluids. Some of the more common process fluids consist of water, liquefied gases and natural gas.

Doppler effect ultrasonic flow measurement

The Doppler effect device relies on objects with varying density in the flow stream to return the ultrasonic energy. With the Doppler effect meter a beam of ultrasonic energy is transmitted diagonally through the pipe. Portions of this ultrasonic energy are reflected back from particles in the stream of varying density. Since the objects are moving, the reflected ultrasonic energy will has a different frequency. The amount of difference between the original and returned signals is proportional to the flow velocity.

General Summary

Most ultrasonic flowmeters are mounted to the outside of the pipe and as such operate without coming in contact with the fluid. Apart from not obstructing the flow, they are not affected by corrosion, erosion or viscosity. Most ultrasonic flowmeters are Bi-directional, and sense flow in either direction.

Advantages

- Suitable for large diameter pipes
- No obstructions, no pressure loss
- No moving parts, long operating life
- Fast response
- Installed on existing installations
- Not affected by fluid properties

Transducer Limitations or Disabilities

- Accuracy is dependant on flow profile
- Fluid must be acoustically transparent
- Errors cause by build up in pipe
- Only possible in limited applications
- Expensive
- Pipeline must be full
- Turbulence or even the swirling of the process fluid can affect the ultrasonic signals
- In typical applications the flow needs to be stable to achieve good flow measurement, and typically this is done by allowing sufficient straight pipe up and downstream of the transducers

- The straight section of pipe upstream would need to be 10 to 20 pipe diameters with the downstream requirement of 5 pipe diameters
- For the transit time meter, the ultrasonic signal is required to traverse across the flow, therefore the liquid must be relatively free of solids and air bubbles
- Anything of a different density (higher or lower) than the process fluid will affect the ultrasonic signal

Summary

Doppler flowmeters are not high accuracy or high performance devices, but do offer an inexpensive form of flow monitoring. Their intended operation is for dirty fluids, and find applications in sewage, sludge and waste water processes.

Being dependent on sound characteristics, ultrasonic devices are dependent on the flow profile, and are also affected by temperature and density changes.

2.10 Level Transmitters

There are numerous ways to measure level that require differing technologies and to encompass all the various units of measurement.

- Ultrasonic, transit time
- Pulse echo
- Pulse radar
- Pressure, hydrostatic
- Weight, strain gauge
- Conductivity
- Capacitive

For continuous measurement, the level is detected and converted into a signal that is proportional to the level. Microprocessor based devices can indicate level or volume.

Different techniques also have different requirements. For example, when detecting the level from the top of a tank, the shape of the tank is required to deduce volume.

When using hydrostatic means, which detects the pressure from the bottom of the tank, then the density is to be known and remain constant.
Level sensing is a simpler concept than most other process variables and allows a very simple form of control. The sensors can be roughly grouped into categories according to the primary level sensing principle involved.

The signals produced by these means must then be converted into a signal suitable for process control applications, such as an electrical, pneumatic or digital signal.

2.10.1 Installation Considerations

The following are outlines of the more important considerations that need to be considered when installing either atmospheric or pressurized vessels.

Atmospheric Vessels

Most instruments involved with level detection can be easily removed from the vessel. Top mounting of the sensing device also eliminates the possibility of process fluid entering the transducer or sensor housing should the nozzle or probe corrode or breaks off. Many level measurement devices have the added advantage that they can be manually gauged. This provides two important factors:

- Measurements are still possible in the event of equipment failure
- Calibration and point checks can provide vital operational information

One common installation criteria for point detection devices is that they be mounted at the actuation level, which may present accessibility problems.

Pressurized Vessels

Two main considerations apply with level measurement devices in pressurized vessels:

- Facilities for removal and installation while the vessel is pressurized.
- The pressure rating of the equipment for the service.

Pressurized vessels can also be used to prevent fugitive emissions, where an inert gas such as hydrogen can be used to pressurize the process materials. Compensation within the level device needs to also be accounted for as the head-pressure changes.

The accuracy of the measuring device may be dependent on the following:

- Gravity variations
- Temperature effects
- Dielectric constant

Also the presence of foam, vapor or accumulated scum on the transducer affects the performance.

Impact on the Overall Control Loop

Level sensing equipment is generally fast responding, and in terms of automated continuous control, does not add much of a lag to the system.

It is good practice though, to include any high and low switch limits into the control system. If the instrumentation does fail or goes out of calibration, then the process information can be acquired from the high and low limits. Apart from the hard-wired safety circuits, it is good practice to incorporate this information into the control system.

Future Technologies

The cost of sensing equipment is not a major consideration compared with the economics of controlling the process. There is therefore a growing demand for accuracy in level measuring equipment.

Newer models incorporate better means of compensation, but not necessarily new technologies. Incorporating a temperature compensation detector in the pressure-sensing diaphragm provides compensation and also an alternative to remote pressure seals and ensures the accuracy and stability of the measurement.

Greater demands in plant efficiency may require an improved accuracy of a device, not just for the actual measurement, but also to increase the range of operation. If the safety limits were set at 90% due to inaccuracies with the sensing device, then an increased range could be achieved by using more accurate equipment.

Demands are also imposed on processes to conform to environmental regulations. Accurate accounting of materials assist in achieving this. Such technologies as RF admittance or ultrasonic minimize the expense of this environmental compliance.

Problems occur in trying to sense level in existing vessels that may be non-metallic. RF flexible cable sensors have an integral ground element which eliminates the need for an external ground reference when using the sensor to measure the level of process materials in non-metallic vessels.

2.11 The Spectrum of User Models in Measuring Transducers

As an example of the extremes that can occur between the same type of measuring transducer, consider the case of the thermocouple. Firstly in its most simple form, it consists of two dissimilar metal wires joined together to form a loop consisting of two junctions or connections. The Seebeck effect. (If the temperature of the two junctions is different, a current will flow in the loop) then comes into play. Looked at in practice, a thermocouple measuring circuit actually measures the ***difference*** between the two junctions forming the circuit.

Unfortunately three major problems occur with this form of temperature measurement:

2.11.1 Voltage Generation of a Thermocouple

Only very low e.m.f's are generated, typically, 1.8 to 6.0 x 10 -12 Volts or so per O°C, so electrically induced noise, either as the normal or common mode type, can become a problem. Normal mode noise being the more difficult one to remove from a system, this usually being achieved by the introduction of guard lead wires, or careful cable screening.

2.11.2 Thermocouple Linearity

The output of any type of thermocouple is not linear relative to the applied or measured temperature range, This can causes linearity, scaling, ranging and calibration problems.

2.11.3 Cold Junction Compensation

To complete any electrical circuit, requires the formation of a loop, so in the case of the thermocouple, as second junction has to exist to achieve this. This is called the "Cold Junction" and usually sits at ambient temperature, which of course varies and introduces measurement errors which can be extremely large, especially if measurement of the physical quantity is close to the ambient or cold junction temperature.

The simplest form of thermocouple application is in the form of a galvanometer which has the sensitivity to measure the low voltages involved. This is equipped with a temperature sensitive compensating resistor, located next to the input terminals where the measuring circuits cold junction is. This resistor forms part of that measuring circuit and corrects the effect, by changing its resistance and hence the current flow in the circuit, of ambient temperature changes.

The problem with this arrangement is that it is direct reading, and hence does not easily lend itself to inclusion in process control systems, and the physical circuit from the indicator to the thermocouple measuring tip has to be "tuned" to a specific resistance for the cold junction compensator to be accurate. To overcome the non-linearity problem, the scale of the instrument is scaled to the "profile" of the related response curve of the thermocouple type being used.

2.12 Instrumentation and Transducer Considerations

There are many considerations that have to be taken into account when selecting instruments and transducers, the following is an explanation and index of the more important aspects of choice.

2.12.1 Signal Transmission Pneumatic versus Electronic

Electronic means for signal transmission and control is becoming more favored, however pneumatic controls are still used and do have advantages in different applications.

Advantages - Electronic

- Lower installation cost
- Lower maintenance
- Higher accuracy (esp. smart instruments)
- Faster dynamic response
- Suitable for long distances
- Digital control system compatible

The primary reason for selecting electronic devices is their compatibility with the control system. With data exchange highways becoming more common it is also easier to obtain more information from the sensor with smart electronics.

Advantages - Pneumatic

- Lower initial hardware cost
- Simple design
- Less affected by corrosive environments
- Easily connected with control valves
- Pneumatics have a prime advantage because of their safety in hazardous locations.

2.12.2 Signal Conditioners

Signal conditioners change or alter signals so that different process devices can effectively and accurately communicate with each other. They are typically used to link process instruments with indicators, recorders, and microprocessor-based control and monitoring systems.
They consist of either:

Signal Conversion

A signal converter is used to change an analog signal from one form to another. This enables equipment with differing signals to communicate.

Signal Boosting

For analog signals (voltage) that are required to be transmitted over long distances, it is possible that the signal may attenuate, or fade. For analog signals (current) in loops that have a number of loop-powered devices, the signal may not be strong enough.

2.12.3 Noise

Electrical noise, or interference, is unwanted electrical signals that cause disruptive errors, or even completely disable electronic control and measuring equipment.

There are two main categories of electrical measurement noise:

1. Radio Frequency Interference (RFI)
2. Electromagnetic Interference (EMI)

Some examples of the more commonly encountered sources of interference are:

- Hand-held (walkie-talkie)
- Cellular phones
- AC and DC motors
- Transformers
- Arc welders
- Large solenoids, contactors and relays
- High power cabling, both voltage and current
- High speed power switching, such as SCR's and thyristor
- Variable frequency drives
- Static discharges
- Induction heating systems
- Radar devices
- Fluorescent lights

Radio Frequency Interference and Electromagnetic Interference can cause unpredictable performance in instrumentation. These types of interference can often be non-repeatable, making it hard to detect, isolate and rectify the problem. RFI and EMI can also degrade an instrument's performance and possible cause the instrument to fail completely.

Any of these problems can result in reduced production rates, process inefficiency, plant shutdowns, and possibly even create dangerous safety hazards.

There are two basic approaches to protecting instrumentation systems from the harmful effects of RFI and EMI.

The first is to keep the interference from entering the system by:

- Shielding
- Proper grounding
- Terminal filters

The second is to design the system so that it is unaffected by RFI and EMI.

Noise Reduction Techniques

Some of the more common techniques for reducing or even eliminating electrically induced noise are:

- **Use of transmitters, i.e. for thermocouples.**

  The signal is more robust to noise over long distances. Typically 4-20mA.

- **Shielded/twisted pair cable.**

  Twisting is done to decouple the wires from induced currents from varying electric and magnetic fields that may exist. The principle of twisting is that equal voltages are induced in each loop of the twisted wires, but of opposite phase which makes them cancel.

- **AC Inductive load circuits**

  For AC inductive loads, use a properly rated MOV across the load in parallel with a series RC snubber.
  An effective RC snubber circuit would consist of a 0.1 uF. capacitor of suitable voltage rating, and a 47 ohm 0.5W resistor.

- **DC Inductive load circuits**

  For DC inductive loads, use of a diode across the load is effective, provided the polarity is correct. Use of an RC snubber circuit can be added as an enhancement.

2.12.4 Materials of Construction

Often when selecting measurement or control equipment, options are available for the various materials of construction. The primary concern is that the process material will not cause deterioration or damage to the device.

Below is a brief list of other qualities or characteristics that assist in the selection of the material of construction.

- 316SS
- Hastelloy C-276
- Monel
- Carbon steel
- Beryllium copper; good elastic qualities
- Ni-Span C ;very low temperature coefficient of elasticity
- Inconel; extreme operating temperatures and corrosive process
- Stainless Steel; extreme operating temperatures and corrosive process
- Quartz; Minimum hysteresis and drift

2.12.5 Signal Linearization

When the output of a device responds at a proportional rate to changes in the input, then the device is linear and there is a constant gain (output/input) over the full range of operation and the resolution remains constant. If the response or reaction of some device in a system is not linear then it may need to be made linear because there are two main problems, when the device is not linear:

- The gain changes
- The resolution and accuracy change

In a control system there are three ways to account for non-linear equipment:

1. Base application on the highest gain
2. Measure the gain at a number of points
3. Modify the gain as a function of the process variable

The simpler way to overcome any non-linearity is to linearize the signal before the control system calculations.

2.13 Selection Criteria and Considerations

Reasons for selecting one type of measuring equipment over another vary, but typically the decisions are based on the perceived advantages and disadvantages of the range of devices available.

A comprehensive list would take into account the following:

- Accuracy
- Reliability
- Purchase price
- Installed cost
- Cost of ownership
- Ease of use
- Process medium, liquid / stem / gas
- Degree of smartness
- Repeatability
- Intrusiveness
- Sizes available
- Maintenance
- Sensitivity to vibration

In addition particular requirements for flow would include:

- Capability of measuring liquid, steam and gas
- Rangeability
- Turndown
- Pressure drop
- Reynolds number
- Upstream and downstream piping requirements

A more systematic approach to selection process measurement equipment would cover the following steps:

2.13.1 Application

This is the requirement and purpose of the measurement.

- Monitor
- Control
- Indicate
- Point or continuous
- Alarm

2.13.2. Processed Material Properties

Many process-measuring devices are limited by the process material that they can measure.

- Solids, liquids, gas or steam
- Conductivity
- Multiphase, liquid / gas ratio
- Viscosity
- Pressure
- Temperature

2.13.3 Performance

This relates to the performance required in the application.

- Range of operation
- Accuracy
- Linearity (Accuracy may include linearity effects)
- Repeatability (Accuracy may include repeatability effects)
- Response time

2.13.4 Installation

Mounting is one of the main concerns, but the installation does involve the access and other environmental concerns.

- Mounting
- Line size
- Vibration
- Access
- Submergence

2.13.5 Economics

The associated costs determine whether the device is within the budget for the application.

- Purchase cost
- Installation cost
- Maintenance cost
- Reliability / replacement cost

2.13.6 Environment and Safety

This relates to the performance of the equipment to maintain the operational specifications, and also failure and redundancy should be considered.

- Process emissions
- Hazardous waste disposal
- Leak potential
- Trigger system shutdown

2.13.7 Measuring Devices and Technology

At this stage the selection criteria is established and weighed up with readily available equipment.
A typical example for flow is shown:

1. DP	-	*Orifice plate*
	-	*Variable area*
2. Velocity	-	*Magmeter*
	-	*Vortex*
	-	*Turbine*
	-	*Propeller*
3. Positive Displacement	-	*Oval gear*
	-	*Rotary*
4. Mass flow	-	*Coriolis*
	-	*Thermal*

2.13.8 Vendor Supply

Limitations may be imposed, particularly with larger companies that have preferred suppliers, in which case the selections may be limited, or the procedure for purchasing new equipment may not warrant the time and effort for the application.

2.14 Introduction to the SMART TRANSMITTER

The most elaborate form of thermocouple transducer, quite often refereed to as a "***Smart transmitter***" as shown in Figure 2.15; comprises:

1. An electronic cold junction compensator
2. A highly stable dc amplifier to get the thermocouples low voltages up to a reasonable operating level
3. Some form of microprocessor that perform a linearization function on the thermocouples generated voltage
4. A ranging function
5. An output or transmitter part of the system, where selectable types of output can be selected
6. And, of course, the thermocouple itself

This then gives the availability to use a thermocouple of a type with a total range of 0°C to 650°C, to be ranged, or set for 100°C to 300°C and this, with a 4 to 20 mA transmitter output, gives an output. = 12.5 °C / mA, linearly through the required or selected range.

In normal use, using the entire range of the thermocouple, we would have an output sensitivity of 650°C / 20-4 mA or 40.625°C / mA giving a sensitivity ratio of 3.25:1

This concept can be applied to most types of measurement transducers, conceptually saying that the output of these devices can be ranged and made linear before being introduced to the controllers input itself.

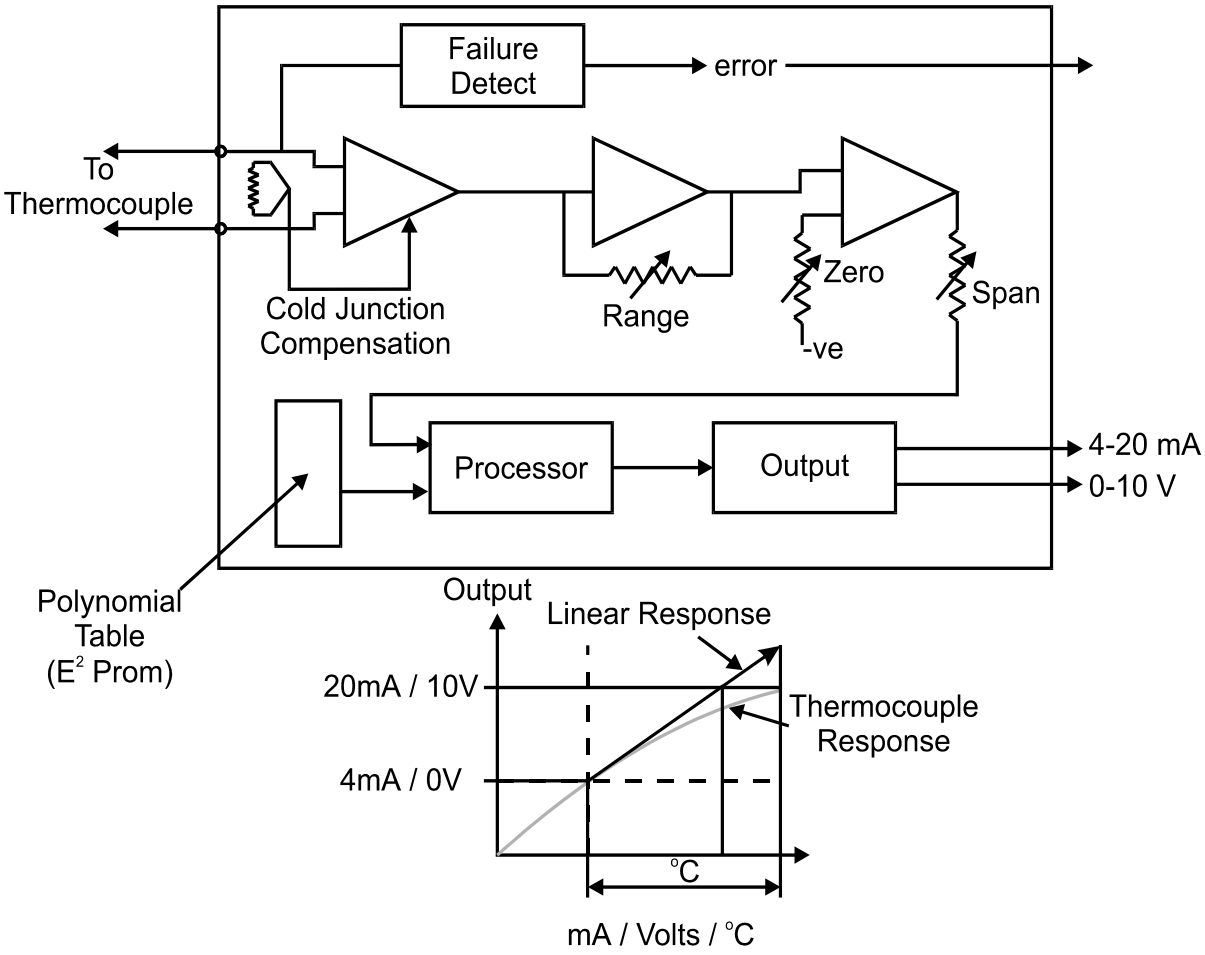


Figure 2.15
Microprocessor Based Thermocouple Measuring System with Ranged and Linear Output